

EE Elective Courses Catalogue 2025



Course Name and Instructor	Description	Expected Learning Outcomes	Hands-On Expectations	Market Demand Fit
EE 416 VLSI Circuits Design Dr. Walid Dyab	This course offers an introduction to Very Large Scale Integration (VLSI) design, exposing students to the complexities of creating integrated circuits with cutting-edge technologies. It begins with an exploration of VLSI systems, outlining key design and fabrication stages, including ASIC and FPGA development and processes like photolithography. Participants will explore significant industry players and evolving market trends and gain practical skills through extensive use of HDLs like VHDL and Verilog for circuit design and simulation. The curriculum also covers functional and timing analysis, post-synthesis optimization, and various packaging technologies. Additionally, the course addresses professional ethics in VLSI design, stressing regulatory adherence, intellectual property integrity, accurate reporting, and environmental awareness, equipping students for the technical and ethical challenges in the semiconductor industry.	1.1 Explain the workflow of VLSI design methodologies and the key steps involved in the fabrication of ASICs and FPGAs. 1.2 Identify the main players in the VLSI industry 2.1 Design VLSI circuits using HDL. 2.2 Perform functional and timing analysis of HDL designs 2.3 Execute post-synthesis analysis to ensure the design meets specified performance criteria. 2.4 Differentiate between different packaging of VLSI circuits 3.1 Uphold ethical standards in VLSI design, including compliance with regulations, accurate reporting, intellectual property rights, and environmental considerations.	AMD/Xilinx Tools (ISE/Vivado) <i>Electric</i>, A software for ASIC Layout Generation Students should be able to design standard cells for CMOS Digital ICs	At least 50 semiconductor design companies will be established in Saudi Arabia by 2030 after a new initiative was launched to develop the sector in the Kingdom. The students will be prepared to join the national hub for semiconductors, and make use of the Saudi Semiconductors Program. Saudi Semiconductors Program (SSP) is a program that targets helping researchers and students in Saudi universities by providing them with the design tools and tapeout areas, and helping them with characterizing the manufactured chips. The program aims at developing human talents that can work in designing electronic chips, and contribute in localizing the semiconductors industry in the Kingdom by removing the hassles in acquiring the necessary resources. https://ssp.kacst.gov.sa/

EE 424 Digital Communications Dr. Walid Dyab/ Dr. Mourad Rizk	This course addresses the problem of designing efficient and reliable digital communication systems through introducing the field of information theory. Source and channel coding which are important elements of information theory are essential for any communication systems. We cover the mathematical preliminaries related to those elements, such as entropy, relative entropy, and mutual information. We show how they arise as natural answers to questions of data compression, channel capacity, and error correction. We also study different source and channel coding schemes, such as Huffman, arithmetic, block, cyclic, and convolution codes.	1.1 Recognize the source and channel coding theorems. 2.1 Explain the notions of entropy, relative entropy and mutual information, and how they are relevant. 2.2 Calculate the capacity of different channels. 2.3 Analyze and design source and channel coding schemes.	Build a complete full duplex digital communication system using software defined radio Simulate wireless communications systems using MATLAB/Simulink	This course equips students with critical skills required in the rapidly evolving field of digital communications. Students will be prepared to tackle real-world challenges across various cutting-edge technologies. The practical focus on widely-used systems like satellite and GPS ensures that graduates are well-versed in the latest industry standards and technologies. This knowledge is important for careers in telecommunications, data storage, network engineering, and beyond. All cellular and networking companies such as STC, Mobily, Zein, Arabsat, Nournet, and many more are looking for candidates and talents who are specialized in communication technologies.
EE 425 Wireless Communications Systems Dr. Walid Dyab / Dr. Maged / Dr. Mourad Rizk	This course explores the technical challenges and solutions in designing the physical and link layers of wireless communication systems. Key topics include wireless channel characterization, cellular network principles, digital modulation methods, spread spectrum, and multiple access techniques such as CDMA and OFDMA. The course also covers diversity techniques and introduces advanced technologies like LTE, MIMO, and 5G NR.	1-Explain fundamental wireless communication principles 2-Describe the challenges in designing the physical and link layers of wireless systems. 3- Apply digital modulation and multiple access techniques 4- Design wireless system components	Simulation using Matlab	The course prepares the students for roles as Wireless Engineer, RF Engineer, Telecom Specialist, and IoT Network Specialist. As industries expand in smart cities, AI-driven networks, and satellite internet (e.g., Starlink, Kuiper), this course ensures graduates meet evolving market needs.

EE 414 Electrical and Electronic Measurements Eng. Sivakumar Selvam / Dr. Mahajan Sagar Bhaskar	The Electrical and Electronics Measurements course covers fundamental principles, instruments, and techniques used to measure electrical and electronic parameters accurately. Students will learn about voltmeters, ammeters, wattmeters, oscilloscopes, and digital measurement systems, along with sensor and transducer applications. Emphasis is placed on error analysis, precision, and modern data acquisition techniques. Through theoretical learning and practical lab sessions, students gain hands-on experience with industry-standard measurement tools, preparing them for real-world applications in electrical and electronic systems.	1. Explain fundamental concepts such as accuracy, precision, sensitivity, and error analysis in electrical and electronic measurements. 2. Demonstrate proficiency in using instruments like voltmeters, ammeters, wattmeters, oscilloscopes, and digital multimeters for accurate measurements. 3. Evaluate measurement data, identify errors, and apply corrective techniques to improve accuracy in electrical and electronic systems. 4. Utilize modern data acquisition systems, sensors, and transducers for real-world applications in power systems, electronics, and automation.	1. Operate and calibrate basic electrical measuring instruments. 2. Perform voltage, current, power, and resistance measurements. 3. Analyze real-time data from sensors and transducers.	The demand for professionals skilled in electrical and electronics measurements is growing across industries such as power systems, automation, telecommunications, and electronics manufacturing. With the rise of smart grids, IoT devices, and advanced automation, accurate measurement and data acquisition are critical for system efficiency and reliability. Companies seek engineers proficient in modern measurement techniques, sensor integration, and data analysis to enhance product quality and performance. This course equips students with essential skills to meet industry needs, improving their employability in sectors that rely on precise electrical and electronic measurements.
EE 442 Digital Signal Processing Sivakumar Selvam / Dr. Mourad Rizk	This course provides an in-depth understanding of Digital Signal Processing (DSP), focusing on the analysis, manipulation, and implementation of digital signals in various engineering applications. Students will explore discrete-time signals and systems, sampling theory, discrete Fourier transform (DFT), fast Fourier transform (FFT), digital filter design, and real-time signal	1. Analyze and manipulate discrete-time signals using time-domain and frequency-domain techniques. 2. Implement Z-transform, DFT, and FFT for efficient signal analysis. 3. Design and implement digital FIR and IIR filters for	-MATLAB Programming – Simulating digital signals, filtering, and spectral analysis. - Filter Design & Optimization – Designing and testing FIR & IIR filters for audio and communication signals.	Digital Signal Processing (DSP) is a fundamental technology driving advancements in telecommunications, embedded systems, biomedical engineering, and radar applications. With the increasing reliance on wireless communication, AI-driven signal processing, and real-time data analysis, professionals with

	processing techniques. The course emphasizes both theoretical concepts and practical implementation, preparing students for advanced topics in signal processing.	various signal processing applications. 4. Apply sampling and reconstruction techniques in real-world scenarios. 5. Utilize MATLAB for signal processing simulation and visualization. Demonstrate the role of DSP in communications, audio processing, and radar systems.	-Speech & Audio Processing – Implementing real-time speech enhancement and noise cancellation.	strong DSP expertise are in high demand. This course equips students with essential skills for designing, analyzing, and optimizing digital systems used in areas such as 5G networks, autonomous systems, speech and image processing, and industrial automation.
EE 454 Power System Planning Sivakumar Selvam / Dr. Umashankar Subramaniam	Pending			
EE 456 Renewable Energy Engineering Sivakumar Selvam/ Dr. Umashankar Subramaniam / Dr. Mahajan Sagar Bhaskar	This course focuses on the principles, technologies, and applications of solar and wind energy systems. It covers energy conversion, system design, efficiency analysis, and integration into power grids. Students will explore key challenges, sustainability considerations, and advancements in photovoltaic (PV) systems and wind turbine technology. The course combines theoretical knowledge with hands-on applications to prepare students for careers in the renewable energy sector.	1. Explain the fundamental principles of solar and wind energy conversion. 2. Analyze the performance, efficiency, and feasibility of photovoltaic (PV) and wind turbine systems. 3. Design and evaluate solar and wind energy systems for various applications. 4. Assess the environmental, economic, and	Conduct experiments on solar panels and wind turbines to measure energy output and efficiency. Use simulation tools to model and optimize solar and wind energy systems. Analyze real-time data from PV systems and wind turbines. Design and implement small-scale solar and wind energy projects.	The demand for skilled professionals in solar and wind energy is rapidly increasing as industries and governments prioritize sustainable power solutions. With the expansion of solar farms, wind parks, and decentralized energy systems, there is a growing need for engineers proficient in system design, energy conversion, and grid integration. This course equips students with the expertise to contribute to the renewable energy transition and meet the industry's evolving needs.

		policy implications of solar and wind energy deployment.		
EE 499 Special Topics in Electrical and Electronic Engineering (Industrial Electric Drives along with Danfoss KSA) Dr. Umashankar Subramaniam / Dr. Mahajan Sagar Bhaskar	This specialized project-based course, cosponsored by Danfoss KSA, provides in-depth knowledge of electrical drives, their fundamental principles, applications, and industry relevance. The course is based on the Danfoss sponsored resources Students will gain theoretical insights along with hands-on experience using Danfoss-sponsored drive equipment. Through this course, students will explore the role of variable frequency drives (VFDs) in modern industrial applications, including energy efficiency, motor control, and smart drive technologies. The curriculum integrates real-world case studies, simulations, and laboratory sessions to enhance practical understanding.	Upon successful completion of the course, students will be able to: Understand Drive Fundamentals – Explain the working principles of electrical drives, including frequency converters, control strategies, and drive components. Analyze Drive Applications – Identify the role of drives in industrial automation, renewable energy systems, and smart grids. Implement Energy-Efficient Solutions – Evaluate and propose drive-based solutions for improving energy efficiency in industrial and commercial settings. Hands-On Operation & Troubleshooting – Gain practical skills in configuring, operating, and troubleshooting Danfoss VFDs.	The course includes extensive hands-on training with Danfoss-sponsored equipment. FC102 & FC 302 Drives and Danfoss online learning portal. Students will: Work with Danfoss VLT® FC 302 and VACON® FC 102 drives to configure and optimize motor control. Conduct real-time parameter tuning and performance testing using frequency converters. Perform fault diagnosis and troubleshooting of industrial drives. Implement drive-based energy-saving techniques in practical scenarios. Utilize software tools (MCT 10 and MCT 30 from Danfoss) for drive	Growing Demand for Drives – With the push for energy efficiency and automation, industries worldwide are adopting VFDs to optimize performance. Industrial Applications – Drives are crucial in manufacturing, Water Desalination, HVAC systems, renewable energy, and smart grids and Aramco Oil refineries Employment Opportunities – Professionals skilled in drive technologies are highly sought after in industries such as automotive, oil & gas, power distribution, and industrial automation. Future-Ready Skills – The course integrates modern trends, including Industry 4.0, IoT-enabled drives, and AI-driven predictive maintenance, preparing students for future advancements in power electronics and automation.

		Explore Advanced Drive Technologies – Learn about modern developments such as predictive maintenance, digital twins, and cloud-based monitoring in drives.	monitoring and control, including cloud-integrated solutions.	
EE 499 Special Topics (optical communication) Dr. Maged	This course focuses on optical communication and networks. It gives the student insight and understanding of the basics of optical networks and the building blocks such as sources, receivers, isolators, and optical filters. The course discusses the types of fiber and the optical signal impairments such as scattering, attenuation, and dispersion. It also covers TDM and WDM optical networks in addition to a brief introduction to wireless optical networks	-Recognize different types of fiber and network components -Explain the optical channel impairments and losses -Learn the principle of operation of optical transmitters and detectors -Familiar with design considerations of fiber optic systems -Demonstrate the ability to work in a team to collaborate in performing design projects	Requires laboratory in optical or paid simulation software	This course equips students with expertise in optical components, signal impairments, and network architectures, preparing them for roles such as Optical Network Engineer, Fiber Optics Specialist, and Telecom Engineer.
EE 426 Antenna and Wave Propagation Dr. Mourad Rizk	This course provides the students with an understanding of the basic principles of Antenna Analysis and Design, an overview of the fundamental characteristics and parameters of antennas, an overview of analytical and numerical methods used to analyze and design antennas with application to some basic antenna structures such as linear antennas, loop antennas, and antenna arrays.	1. Describe radiation mechanism, i.e. how electromagnetic fields are generated, guided, and radiated. 2. Recognize some advanced antennas types. 3. Explain the various types of antennas and their use in wireless communication. 4. Analyze the effect of the antenna arrays on performance. 5. Calculate the antenna parameters like gain, input	HFSS Antenna simulation MATLAB simulation 3D antenna printing	The demand for antenna engineering expertise is rapidly increasing with the expansion of 5G networks, satellite communications, IoT, and autonomous systems. Industries such as telecommunications, aerospace, automotive, and defense require skilled engineers to design and optimize high-performance antennas for modern applications. This antenna course is designed to equip students with both theoretical knowledge and hands-on skills in

		impedance, radiated power and radiation resistance. 6. Design antennas for various applications.		antenna design, testing, and optimization. It covers key areas such as antenna theory, and simulation tools (HFSS) to align with industry demands. Graduates will be well-prepared for careers in wireless communications, radar systems, and emerging 6G technologies, making them valuable assets in today's technology-driven job market.
EE 415 Opto-Electronics Dr Maged	Overview of semiconductor fundamentals; Heterojunction structures; Light-emitting devices: principles, spectral characteristics, and modulation bandwidth; Photodetectors: operational principles, noise analysis, and sensitivity; Optical waveguides; Fundamentals of optical fibers, including attenuation and dispersion; Optical amplification technologies; WDM and DWDM systems along with their components; Key elements of optical networking; Emerging trends and advancements in optoelectronics.	-Understand light sources and detectors -Design and evaluate optical network solutions -Analyze optical signal impairments	Requires laboratory	This course prepares students for careers in telecommunications, semiconductor manufacturing, biomedical optics, and photonic circuit design, with roles such as Optoelectronics Engineer, Photonics Specialist, and Optical Systems Designer. As industries expand in 5G, LiDAR, quantum optics, and AI-driven optical computing, this course ensures graduates are equipped with the skills needed for cutting-edge innovations.
EE 417 Communications Electronics	Pending			
EE 423 Data Communication Networks Dr Muddesar	This course provides a comprehensive understanding of data communication networks with a strong focus on data communication and networking techniques for smart system applications such smart grids, energy systems for smart homes, buildings and factories. It covers network	1) Understand basic data communication and network concepts and protocols 2) Implement and evaluate middle-layer data access protocols	1)Configuring and implementing Data Communication Network using Raspberry Pi/Arduino.	The demand for networking professionals is growing rapidly due to the increasing reliance on digital communication. With advancements in cloud computing, cybersecurity, and IoT, networking skills are essential for various roles in IT infrastructure,

	architectures, protocols, and data transmission principles, emphasizing wireless and wired communication technologies essential for Smart systems applications. The course covers several protocols operating at different layers of OSI and TCP/IP models. The course will focus on implementing Middle layer data access layer protocols (MQTT/CoAP etc), to provide real-time data communication, and networking for IoT-enabled smart systems. The course also explores Smart System integration with cloud-based application layers. It also covers data communication fundamentals, such as modulation, error correction, and synchronization techniques relevant to real-time Smart Systems applications.	3)Integrate Smart Systems with cloud-based networking solutions	2)Implementation of Middle layer communication protocols (MQTT, CoAP) to enable real-time data exchange among multiple microprocessors and microcontrollers to build prototypes for smart grids, energy systems, and industrial IoT applications. 3) Deploying and integrating smart systems with cloud-based platforms for remote monitoring, data analytics, and automation. 4) Developing and testing secure data transmission techniques	cybersecurity, & telecommunications. Companies such as STC, Mobily, and other telecom and IT firms in Saudi Arabia and worldwide seek candidates with foundational networking knowledge. This course prepares students for entry-level positions in networking and sets the stage for obtaining industry certifications such as CCNA (Cisco Certified Network Associate).
EE 433 Mechatronics Dr Mahmoud Abdulraheem	This course provides a multidisciplinary introduction to Mechatronics, focusing on the synergetic integration of mechanical systems, electronics, sensors, actuators, microcontrollers, and information technology to create intelligent automation solutions. It starts with thinking how the required function can be realized by the combination of	1.1 Understand the concept of mechatronic system design and how to construct a conceptual framework to select an appropriate solution approach 2.1 Learn how to work with sensors, actuators, controllers and how to read and interpret data sheets.	Working with different sensors, actuators and controllers. 3D modeling of mechanical systems using SolidWorks. Modeling and simulation of control systems using MATLAB and LabVIEW.	Mechatronics engineers develop and design mechanical products with electronic control technology using electronic components (microcomputers). Unlike devices in the past which operated solely on mechanics, mechatronics engineering has enabled devices to be controlled by software. Many products, such as motor vehicles, homeware appliances,

	different subsystems according to a Systems Engineering approach (V-model). Students will explore electro-mechanical systems controlled by microcontrollers, gaining hands-on experience in instrumentation, measurement systems, data acquisition, and control techniques. Through practical applications, students will learn how to design, implement, and troubleshoot sensor-actuator systems, interface microcontrollers with real-world mechanisms, and develop automated control solutions.	2.2 Learn how to model and simulate mechatronic systems and how to verify design solutions before their implementation. 3.1 Write technical reports following a well-defined design procedure	Design project for a selected mechatronic system.	medical devices and industrial machinery are dependent on mechatronics technology, which increases the value of its technology. Moreover, the popularization of IoT has further pushed the rise in demand for mechatronics engineering. A majority of roles for mechatronics engineers are centred around manufacturing industries such as automobile manufacturers and electrical manufacturers. In particular, the automobile industry is experiencing a shift in demand for mechatronics engineers and electrical engineers from mechanical engineers due to the development of autonomous driving technology and electric vehicles.
EE 434 PLC Microcontrollers Dr Mahmoud Abdulraheem	This course provides a comprehensive introduction to Programmable Logic Controllers (PLCs), equipping students with both theoretical knowledge and hands-on experience in industrial automation. Students will explore the core principles of PLC operation, including relay ladder logic programming, and gain practical exposure to real-world industrial applications. Through interactive lab exercises and case studies, students will learn about the structure and functionality of PLCs, covering	1.1 Understand the main architecture of PLC controllers and their principle of operation 1.2 Learn the basics of programmable logic controllers 1.3 Classify input and output interfacing devices with PLC 2.1 Write PLC programs for Schneider PLC Modicon M241	The course offers extensive hands-on training with Schneider Electric PLCs (TM241 and TM262), Human-Machine Interfaces (HMI), and Schneider's industry-leading software for PLC and HMI programming. Students will gain practical experience in designing, programming, and troubleshooting industrial	The demand for industrial automation and smart manufacturing is surging, fueled by the advancements of Industry 4.0, digital transformation, and the widespread adoption of PLCs across various industries. As automation becomes the backbone of modern industry, companies are increasingly seeking skilled professionals who can design, program, and maintain PLC-based control systems.

	I/O modules, relays, counters, timers, and data manipulation techniques. Additionally, students will examine how PLCs integrate into modern automated manufacturing systems and explore their role in industrial networking and smart factory solutions.	2.2 Execute, debug and test the programs developed for digital and analog operations. 2.3 Determine I/O requirement of application, develop program and interface system with PLC solved using different approaches	automation systems through real-world applications. In addition to working with physical PLC hardware, students will explore virtual industrial automation using Factory I/O, an interactive 3D simulation software. They will build and automate industrial processes, integrating them with Schneider PLCs via Advantech data acquisition systems, bridging the gap between simulation and real-world implementation. By the end of the course, students will have developed essential automation skills, enabling them to confidently program PLCs, configure HMIs, and implement industrial control systems in smart factories and manufacturing environments.	This course is designed to bridge the gap between academia and industry, equipping students with the hands-on expertise and technical knowledge required to thrive in the fast-evolving automation sector. Industries such as manufacturing, oil & gas, power generation, automotive, and logistics rely heavily on PLC-controlled systems to optimize production, enhance efficiency, and ensure operational safety. As a result, PLC programming and automation skills are among the most sought-after competencies in today's job market
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EE 435 Robotics and Automation Dr Muddesar	This course has basic Python programming as a prerequisite. The course delivers fundamental concepts of robotics and automation, designed for electrical engineering students those has basic Python knowledge. The course covers, topics those are essentials to build robots, including actuators, sensors, robot kinematics, control systems, and automation technologies. Furthermore, the course will provide insights into robot motion, position and force control, and mobile robotics. The course will also explores robotic operating systems (ROS), industrial robotics, vision systems, and automation techniques used in manufacturing and smart systems. The course is designed to be applied in nature, enabling students to use python to programming automation and robotics sensors, motion planning, and AI-driven control mechanisms, preparing them for real-world applications in robotics and industrial automation.	1) Understand the essential concepts of robotics and automation, including sensors, actuators, and robot kinematics. 2) Develop and analyze motion planning and control strategies for robotic systems. 2) Implement robot programming using Python and Robot Operating System (ROS). 3) Implement python libraries for machine vision, artificial intelligence, and automation techniques in robotic applications. 3) Design and implement robotic solutions for industrial, healthcare, and smart infrastructure applications.	1) Programming robotic systems using Python and Robot Operating System (ROS). 2) Implementing motion planning and control strategies for robotic manipulators and mobile robots. 3) Designing and testing automation solutions for industrial and manufacturing applications. 4) Integrating sensors and actuators for real-time robotic perception and decision-making.	The demand for robotics and automation professionals is rapidly increasing as industries transition toward smart manufacturing, autonomous systems, and intelligent automation. Robotics plays a critical role in automotive, healthcare, logistics, and renewable energy sectors, where automation enhances efficiency and reduces operational costs. With the rise of Industry 4.0, AI-driven robotics, and autonomous systems, skilled professionals are needed to develop and maintain robotic systems, vision-based automation, and intelligent control mechanisms. Governments and industries worldwide are investing in robotics-driven infrastructure, smart cities, and automated healthcare solutions, creating a growing demand for engineers proficient in robotics and automation.
EE 436 Artificial Intelligence Dr Muddesar	This course provides an introduction of Artificial Intelligence (AI) for Electrical Engineering students. This AI course covers basic AI concepts, including problem-solving techniques, search algorithms, machine learning, and knowledge representation, where the main focus will be on practical applications in electrical engineering.	1) Understand the basics of AI, and AI Concepts including search strategies, machine learning, and knowledge representation. 2) Develop AI-based solutions for electrical engineering applications, such as power	1) Learn about Algorithms, Implementation of AI algorithms using Python libraries like TensorFlow, Numpy and Scikit-Learn. 2) Learn about developing and training machine learning models for real-	AI is one of the fastest growing industries. The growing reliance on AI based predictive maintenance, energy optimization, and intelligent control systems has led to an increased demand for engineers with AI knowledge. Various companies are integrating AI to enhance efficiency and performance. Governments and

	The course integrates hands-on experience with Python-based AI tools such as NumPy, Pandas, Scikit-Learn, and TensorFlow. Specific applications include load forecasting in power systems, AI-based signal processing, and control system optimization. By the end of the course, students will learn about theoretical knowledge and practical skills to apply AI techniques to real-world electrical engineering problems, preparing them for advanced AI applications and research in intelligent automation, smart grids, and predictive maintenance.	system optimization and signal processing. 3) Learn about AI libraries for data preprocessing, model training, and evaluation. 4) Analyze and evaluate AI models using statistical and mathematical techniques. 5) Understand the future implications of AI in engineering disciplines.	world applications in electrical engineering. 3) Work with datasets for training AI models 4) Design and evaluate a small-scale AI project focused on solving an electrical engineering problem.	industries worldwide are investing in AI-based infrastructure and smart city projects, further increasing the need for professionals skilled in AI applications. This course equips students with essential AI skills, positioning them for high-demand careers in AI-based electrical engineering sectors such as power systems, automation, IoT, and robotics
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